

Self-Commissioning Drive for PMSM using CCS-MPC with prognostics using Artificial Intelligence

Zeroth, First & Second Review – Final Year Project

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1. Broad Area of Project Work

- **Power Electronics and Electric Drives** – control of Permanent Magnet Synchronous Motors (PMSM)
- **Model Predictive Control (MPC)** based current control schemes – specifically **Continuous Control Set Model Predictive Control (CCS-MPC)**
- **Self-commissioning / auto-tuning drive systems** – automatic estimation of motor parameters without manual intervention
- **Target application domains:** Electric Vehicle traction drives, industrial servo drives, robotics actuators
- Intersection of *control theory*, *power electronics*, and *embedded systems*

2. Motivation – Why Does This Problem Matter?

- Standard PI-based FOC relies on linear controllers that struggle to maintain optimal dynamic response, decoupling, and strict constraint handling across the highly non-linear, multi-variable operation.
- Conventional PMSM drives rely heavily on cascaded PI controllers – gains must be **manually tuned** for every individual motor's electrical and mechanical traits
- Manual commissioning is **time-consuming**, requires skilled personnel, and increases factory/EV line downtime
- CCS-MPC gives superior dynamic response, but its inner current prediction model is **highly sensitive** to accurate motor parameters (R_s , L_d , L_q , λ_f)
- A drive that can **self-identify its own parameters** can auto-tune both its outer speed loop (using J , B) and inner MPC loop, enabling true **plug-and-play** operation
- Strong industry pull from EV and automation sectors for **adaptive, parameter-robust** drive electronics that also flag their own incipient inverter faults

3. The Questions We Must Answer

- ❶ **Parameter Estimation:** How can the electrical (R_s, L_d, L_q, λ_f) and mechanical (J, B) parameters of a PMSM be accurately estimated at standstill/low-speed to eliminate manual tuning?
- ❷ **MPC Integration:** How can these identified parameters be directly embedded into a CCS-MPC current predictor (inner loop) and PI speed controller (outer loop) to achieve plug-and-play operation?
- ❸ **Control Robustness:** What is the sensitivity of the CCS-MPC prediction accuracy and closed-loop performance to parameter estimation errors, and how can robust tracking be maintained?
- ❹ **Early Fault Detection:** Can an AI-based prognostic module reliably detect incipient inverter switch-level faults early within a Hardware-in-the-Loop (HIL) environment by reusing the existing current and voltage sensing infrastructure?

4. Literature Review (1/3)

Paper Title	Authors	Method Adopted
(Base Paper) On Continuous-Set Model Predictive Control of Permanent Magnet Synchronous Machines (2022)	I. Hammoud, et al.	Realizes a real-time CCS-MPC current controller by solving a constrained optimization problem online.
FPGA Implementation of AI-Based Inverter IGBT Open Circuit Fault Diagnosis (2022)	N. Rajeswaran, et al.	Implements an AI-based condition monitoring system directly onto an FPGA using neuro-genetic algorithms.
Advanced Deep Learning Approaches for Fault Detection in Inverter-Driven PMSMs (2024)	A. Bacha, et al.	Integrates transformer architectures with Physics-Informed Neural Networks (PINNs) and sensor fusion.
Continuous Control Set Predictive Control with Affine Registration Technique (2024)	W. Zhao, et al.	Proposes a CCS-MPC framework combined with an affine registration technique to mitigate parameter sensitivity.

4. Literature Review (2/3)

Paper Title	Authors	Method Adopted
A Comprehensive Review of Parameter Estimation Techniques for PMSM (2020)	M. S. Rafaq, J. Jung	Reviews offline/online parameter estimation techniques to identify methods with fast, accurate convergence.
Global Identification of Electrical and Mechanical Parameters in PMSM Drive (2018)	Z. Liu, et al.	Utilizes a dynamic self-learning Particle Swarm Optimization (PSO) to map and estimate parameters.
Real-Time Feasibility of Data-Driven Predictive Control for Synchronous Motor Drives (2023)	P. Carlet, et al.	Introduces a data-enabled predictive control algorithm to reduce computational complexity for embedded systems.
Weight-Adaptable Disturbance Observer for CCS-MPC of NPC-3L-Fed PMSMS (2025)	Z. Liang, et al.	Implements a cascaded CCS-MPC strategy with an on-line weight-adaptable disturbance observer.

4. Literature Review (3/3)

Paper Title	Authors	Method Adopted
An open-circuit fault diagnosis method for PMSM drive system based on multi-branch 1D-CNN (2024)	Y. Zhang, et al.	Utilizes a 1D-CNN to directly extract fault features from raw 3-phase current signals without manual engineering.
Mechanism-Based Fault Diagnosis Deep Learning Method for PMSM (2024)	L. Li, et al.	Combines Continuous Wavelet Transform (CWT) with a CNN model to classify specific faults physically.

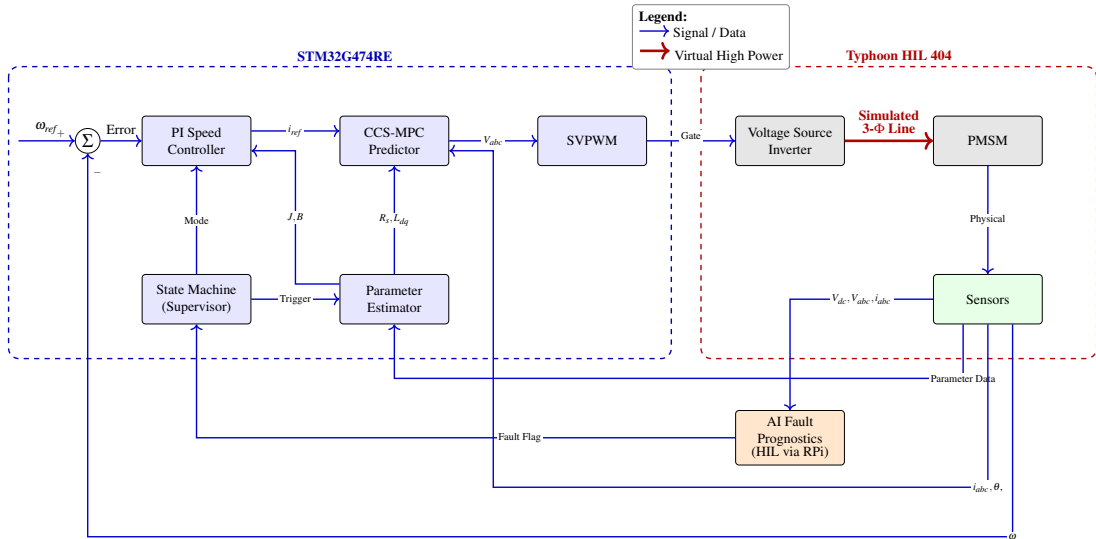
Research Gap

While isolated solutions exist for automated parameter identification, MPC robustness, and AI-based fault diagnostics, there is a lack of a unified framework. Most existing work corrects parameters *during operation* rather than fully self-commissioning prior to standard running, and rarely integrates the drive's data layer simultaneously for **Dual-Loop Auto-tuning** and **AI-based inverter fault prognostics**.

5. Our Solution – Specific Objectives

- **Objective 1 (Self-Commissioning Simulation):** Design and simulate a self-commissioning CCS-MPC PMSM drive – standstill DC/AC pulse tests for electrical parameters, and a brief low-speed test for mechanical parameters (J, B). Feed J, B into an auto-tuned PI speed loop, and continuously update the CCS-MPC inner state-space matrices.
- **Objective 2 (MIL & SIL Verification):** Verify the control algorithm through a structured Model-in-the-Loop (MIL) and Software-in-the-Loop (SIL) testing pipeline, utilizing Model-Based Design to auto-generate production C-code for the target microcontroller.
- **Objective 3 (HIL Real-Time Validation):** Validate real-time control performance via Hardware-in-the-Loop (HIL) testing. Execute generated code on the physical STM32G474RE while interfacing with a real-time plant emulator simulating the PMSM and inverter power stage.
- **Objective 4 (AI Prognostics Deployment):** Integrate an AI-based fault prognostic module deployed on a Raspberry Pi. Inject representative inverter switch-level faults within the HIL setup to evaluate real-time signature analysis and anomaly detection.

6. Proposed System – The HIL Architecture



7. Methodology – The Simulation-to-HIL Pipeline

- **O1 – Simulation & Estimation Setup:** Model PMSM in MATLAB/Simulink. Implement sequences to gather electrical and mechanical parameters dynamically, creating the baseline logic for the control loops.
- **O2 – MIL & SIL Verification:**
 - **Model-in-the-Loop (MIL):** Validate the dual-loop auto-tuning control algorithm against continuous analytical plant models.
 - **Software-in-the-Loop (SIL):** Utilize Embedded Coder to auto-generate C-code from the control logic. Execute this compiled code on the host PC to ensure algorithmic integrity before hardware deployment.
- **O3 – HIL Validation:** Interface the physical **STM32G474RE microcontroller** with the real-time plant emulator. The MCU executes the CCS-MPC and SVPWM natively, passing hardware gate signals to the simulator and receiving analog/digital feedback in real-time.
- **O4 – Prognostics Deployment:** Inject controlled inverter switch-level faults via the HIL software interface. Utilize a **Raspberry Pi** to intercept the real-time analog signatures and execute the AI fault classification model, completing the diagnostic loop.

8. Parameter Identification & Auto-Tuning Strategy

1. Electrical Parameter Estimation (Standstill)

- **Stator Resistance (R_s):** Apply a DC voltage pulse across the phases and measure the resulting steady-state current (V/I).
- **Inductances (L_d, L_q):** Inject high-frequency voltage pulses or step voltages to analyze the current rate-of-change (di/dt) along the d and q axes.

2. Mechanical Parameter & Flux Estimation Sequence

- **Step 1 - Flux Test (λ_f):** The drive spins the motor up open-loop, cuts the power, and measures the unpowered Back-EMF voltage coast-down to calculate λ_f .
- **Step 2 - Friction Test (B):** Now that it knows λ_f , it spins the motor back up to a constant speed. It measures i_q , calculates the exact T_e it is generating, and divides that by the speed to find B .
- **Step 3 - Inertia Test (J):** From that constant speed, it cuts the torque to zero and watches how fast the motor decelerates to find J .

3. PI Speed Loop Auto-Tuning

- The identified parameters dictate the outer speed controller tuning based on a desired closed-loop bandwidth (ω_c).
- The PI gains are calculated.

10. Supervisory Control: State Machine, Mode & Trigger

1. The State Machine Sequence

- **State 0 (IDLE):** Waiting for start command, switches are OFF.
- **State 1 (STANDSTILL):** Injecting DC/AC pulses to find R_s and L_{dq} .
- **State 2 (ALIGNMENT):** Locking the rotor to $\theta = 0$.
- **State 3 (OPEN-LOOP SPINUP):** Using Openloop control to reach steady speed.
- **State 4 (COAST-DOWN):** Switches OFF, measuring Back-EMF and deceleration.
- **State 5 (CALCULATE GAINS):** Computing K_p, K_i and MPC matrices A_d, B_d .
- **State 6 (NORMAL RUN):** Closing the loops, standard MPC drive operation.

2. Mode Signal (To PI Controller)

- **DISABLED:** Outputs zero (during standstill & coast-down).
- **OVERRIDE:** Passes hardcoded open-loop torque limit (during spinup).
- **ACTIVE:** Uses auto-tuned gains.

3. Trigger Signal (To Estimator)

- **Capture R_s, L_{dq} :** Triggered instantly during State 1 pulses.
- **Capture λ_f, J, B :** Triggered at exact moments in States 3 & 4 to sync the math calculation with physical events.

11. Expected Outcome – What Success Looks Like

- A working **self-commissioning algorithm** that estimates both electrical (R_s, L_{dq}, λ_f) and mechanical (J, B) parameters within an acceptable error margin
- An **Auto-tuned PI outer loop** and a **CCS-MPC inner current loop** (on STM32G474RE) that dynamically update their internal matrices based purely on self-identified data
- Demonstrated **reduction in commissioning time** compared to conventional manual-tuning workflows
- A comprehensive **Hardware-in-the-Loop testbed** demonstrating adaptive motor control interacting safely with high-fidelity simulated power stages
- An **AI-based prognostic module** capable of early detection of inverter switch-level (SVPWM) faults, fully deployed and validated on a **Raspberry Pi** during HIL fault-injection testing

12. Software and Hardware Requirements

Software Toolchain

- MATLAB / Simulink
- Simscape Electrical (for Plant Modeling)
- Stateflow (supervisory test logic)
- Embedded Coder (C-code generation)
- STM32CubeIDE (for MPC control flash)
- Python / TensorFlow (for AI prognostics on Raspberry Pi)

Hardware & Emulation

- **Microcontroller (STM32G474RE)** for real-time cascaded control execution
- **Raspberry Pi** (for AI Prognostics deployment)
- Real-time HIL target (Typhoon 404 HIL)
- Interface logic shifters & jumper cabling
- Oscilloscope / Logic Analyzer

13. Cost Estimate of the Project

Item	Purpose	Approx. Cost (INR)
Microcontroller dev kit (STM32G474RE)	Real-time embedded control	3,000 – 5,000
Raspberry Pi kit	AI Prognostics Deployment	4,000 – 7,000
Interface Components	I/O shifting for HIL connection	1,000 – 2,000
HIL platform usage / lab access	Real-time plant emulation	As per lab availability
Estimated Total		INR 8,000 – 14,000

**Note: By focusing on HIL validation rather than a physical high-voltage power stage, prototype costs and hardware risk are significantly minimized while preserving control complexity.*

14. Time Frame for Execution (8 Months)

Task	M1	M2	M3	M4	M5	M6	M7	M8
Lit. survey & topic finalization	■							
Standstill ID & Low-Speed ID logic	■	■						
CCS-MPC integration & simulation		■	■					
MIL testing & PI Auto-tuning			■	■				
SIL testing (C-code generation)			■	■	■			
HIL setup & STM32 interfacing				■	■	■		
HIL validation of Auto-tuning drive					■	■	■	
HIL Inverter Fault Injection sequencing						■	■	■
AI Prognostics Training with Raspberry Pi & Integration						■	■	■
Final HIL System Demo							■	■
Documentation & final review								■
Semester 1: M1–M4 (Algorithm + MIL + SIL) Semester 2: M5–M8 (HIL Interfacing + AI Prognostics + Final Validation)								

15. References (IEEE Format)

Base Paper: I. Hammoud *et al.*, “On Continuous-Set Model Predictive Control of Permanent Magnet Synchronous Machines,” *IEEE Trans. Power Electron.*, vol. 37, no. 9, Sept. 2022.



M. S. Rafaq and J. Jung, “A Comprehensive Review of State-of-the-Art Parameter Estimation Techniques for PMSM,” *IEEE Trans. Ind. Inform.*, vol. 16, no. 7, 2020.



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A. Bacha *et al.*, “Advanced Deep Learning Approaches for Fault Detection and Diagnosis in Inverter-Driven PMSM Systems,” *IJACSA*, vol. 15, no. 12, 2024.



W. Zhao *et al.*, “Continuous Control Set Predictive Control with Affine Registration Technique for PMSM Drive,” *Energies*, vol. 17, no. 18, 2024.



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P. Carlet *et al.*, “Real-Time Feasibility of Data-Driven Predictive Control for Synchronous Motor Drives,” *IEEE Trans. Power Electron.*, vol. 38, no. 2, 2023.



Z. Liang *et al.*, “Weight-Adaptable Disturbance Observer for CCS-MPC of NPC-3L-Fed PMSMS,” *Energies*, vol. 18, no. 21, 2025.



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L. Li *et al.*, “Mechanism-Based Fault Diagnosis Deep Learning Method for PMSM,” *Sensors*, vol. 24, no. 19, 2024.



M. Gierczyński and L. M. Grzesiak, “Comparative Analysis of the Steady-State Model Including Non-Linear Flux Linkage Surfaces and the Simplified Linearized Model when Applied to a Highly-Saturated Permanent Magnet Synchronous Machine—Evaluation Based on the Example of the BMW i3 Traction Motor,” *Energies*, vol. 14, no. 9, art. 2343, 2021.

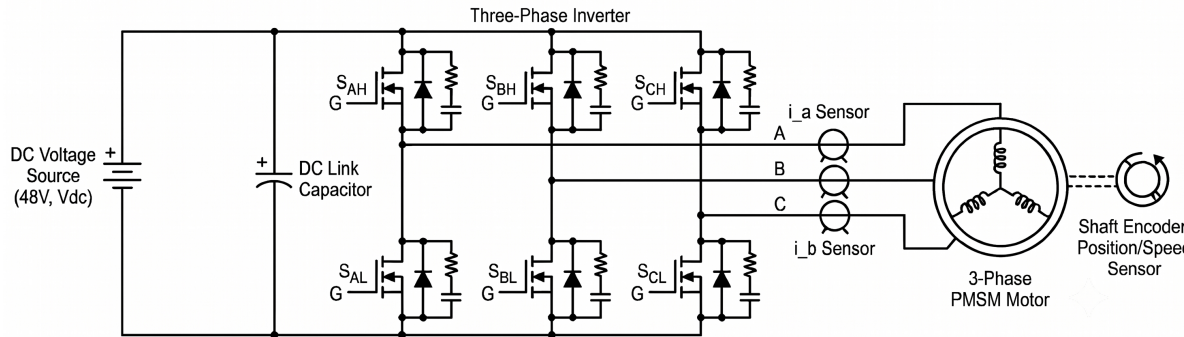
First Review

Progress & Architecture Analysis

16. The Industry Problem & Our Solution

- **The Industry Standard:** Permanent Magnet Synchronous Motors (PMSMs) dominate EV traction and industrial automation due to high efficiency. However, their drives rely almost entirely on Field-Oriented Control (FOC) governed by manually tuned PI controllers.
- **The Bottleneck:** Whenever a motor is swapped on an assembly line, technicians must halt production to manually re-tune the K_p and K_i gains to match the new motor's specific electrical and mechanical traits.
- **Why CCS-MPC for PMSM:** PMSMs exhibit highly coupled, non-linear d-q axis dynamics (especially at high speeds). CCS-MPC explicitly handles these cross-couplings and multi-variable constraints analytically, enabling superior transient response without cascaded PI tuning.
- **Our Proposed Solution:**
 - ① Replace the rigid FOC current loops with a **Continuous Control Set Model Predictive Control (CCS-MPC)** algorithm that analytically calculates optimal voltages.
 - ② Introduce an **AutoTuner State Machine** to autonomously execute pre-operational self-commissioning (extracting $R_s, L_d, L_q, \Psi_{PM}, J, B$) and directly feed them to the MPC, enabling true plug-and-play capability.

16a. Plant Model: Power Stage



16b. System Parameters (Plant & Controller)

PMSM & Inverter Parameters

Standard Application: BMW i3 Traction Motor (EV) [10]

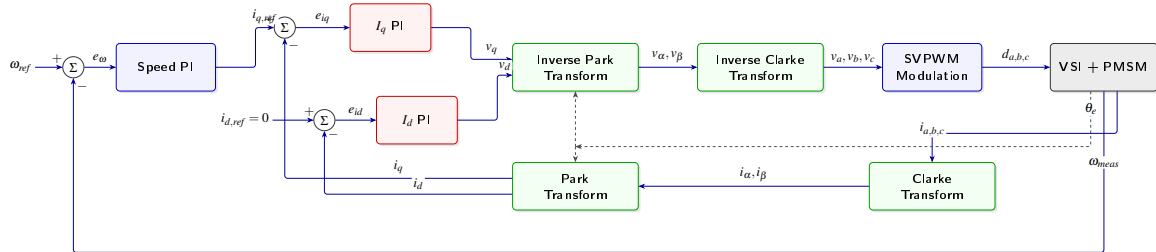
Parameter	Symbol	Value
Nominal DC Bus Voltage	V_{dc}	275.7 V
MOSFET On-Resistance	R_{on}	1.0 m Ω
Switching Frequency	f_{sw}	10 kHz
Stator Resistance	R_s	5.3 m Ω
d/q-axis Inductance	L_d, L_q	71.2 μ H, 141.3 μ H
PM Flux Linkage	Ψ_{PM}	0.0436 Wb
Pole Pairs	p	6
Rotor Inertia	J	0.0867 kg·m ²
Viscous Friction	B_{fric}	0.001 N·m·s/rad
Torque Constant	K_t	0.392 N·m/A
Peak Torque	T_{rated}	258.2 N·m

Control Loop & Simulation

Parameter	Symbol	Value
Control Sample Time	$T_{s,ctrl}$	100 μ s
Simulation Time Step	$T_{s,sim}$	5 μ s
Speed PI Gains	$K_{p,spd}, K_{i,spd}$	2.0, 10.0
Current PI Bandwidth	$\omega_{c,curr}$	$2\pi \times 1000$ rad/s
Max Phase Current	I_{peak}	565.7 A
CCS-MPC Error Weights	Q_{id}, Q_{iq}	1.0, 1.0
CCS-MPC Switch Weight	R_{sw}	0.001
Max Phase Voltage	$V_{ph,max}$	159.2 V
Inverter Deadtime	T_{dead}	2 μ s

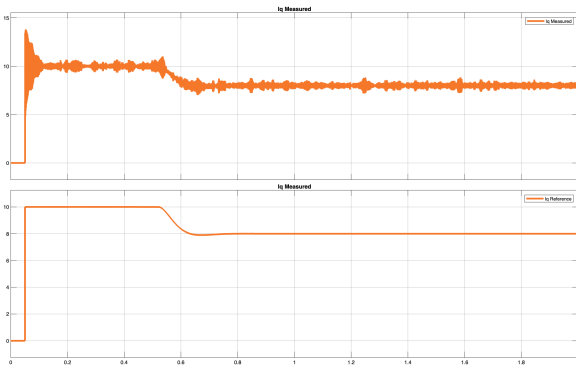
Note: Parameters verified via Comparative Analysis of Steady-State Models [6].

17. Conventional Field-Oriented Control (FOC) Architecture

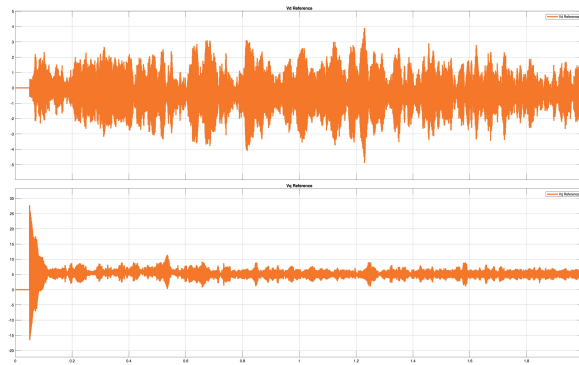


17a. FOC Simulation Results – Constant Speed: Current Tracking

I_q Tracking



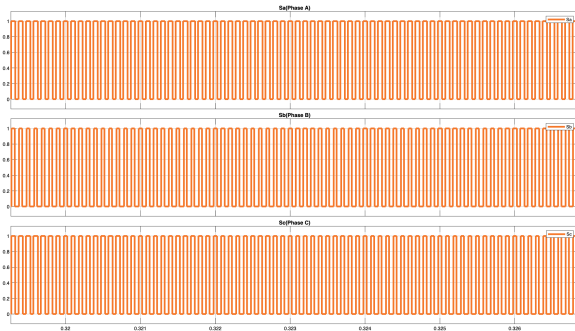
V_d, V_q Tracking



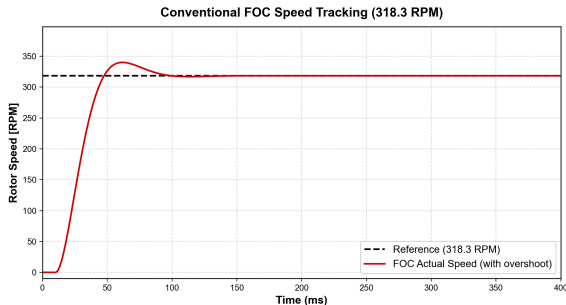
- **Current Tracking Dynamics:** FOC current tracking under constant speed reference shows visible ripple on both axes due to cascaded PI loop delays and cross-axis coupling.

17b. FOC Simulation Results – Constant Speed: Switching & Speed

Switching Pulses (S_a, S_b, S_c)

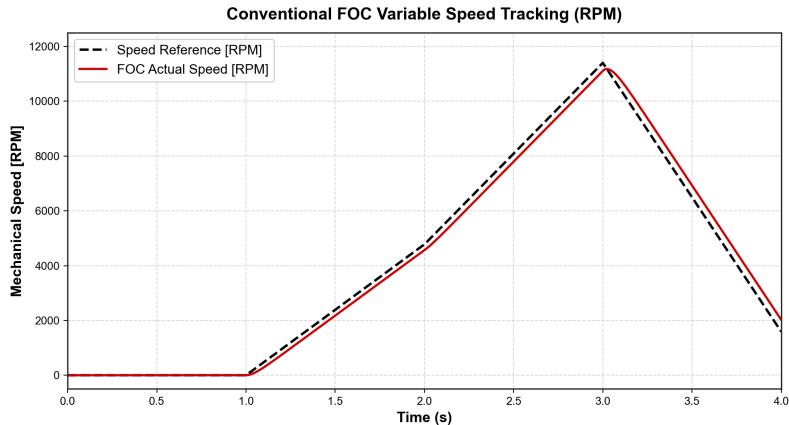


Speed Tracking



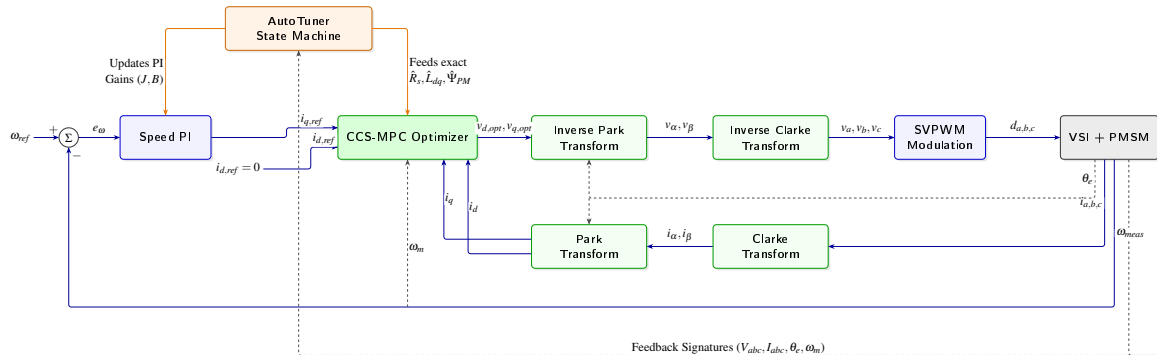
- **Speed Response:** The FOC speed response settles around the 318.3 RPM reference (200 rad/s electrical) after an initial overshoot, with steady switching action from the SVPWM stage.

17c. FOC Simulation Results – Variable Speed Tracking



- **Tracking Lag:** Under multi-step variable speed profiles (ramp – hold – ramp), the FOC speed loop exhibits visible lag during transitions and increased ripple as speed rises.

18. Our Solution: CCS-MPC Architecture



19. CCS-MPC: Predictive Model & Cost Function

Predictive Model (Forward Euler Discretization):

$$i_d(k+1) = i_d(k) + \frac{T_s}{L_d} (v_d(k) - R_s i_d(k) + \omega_e(k) L_q i_q(k)) \quad (1)$$

$$i_q(k+1) = i_q(k) + \frac{T_s}{L_q} (v_q(k) - R_s i_q(k) - \omega_e(k) (L_d i_d(k) + \Psi_{PM})) \quad (2)$$

Deriving Optimal Voltages ($v_{d,opt}$ & $v_{q,opt}$):

❶ **Define Unconstrained Cost Function:**

$$J_{cost} = (i_{d,ref} - i_d(k+1))^2 + (i_{q,ref} - i_q(k+1))^2 \quad (3)$$

❷ **Analytic Minimization:** Differentiate cost function and equate to zero $\left(\frac{\partial J_{cost}}{\partial v_{dq}} = 0 \right)$.

❸ **Extract Closed-Form Solution:** Solve for the exact voltages:

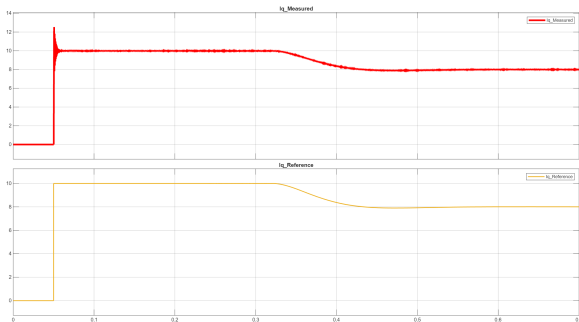
$$v_{d,opt} = \frac{L_d}{T_s} (i_{d,ref} - i_d(k)) + R_s i_d(k) - \omega_e L_q i_q(k) \quad (4)$$

$$v_{q,opt} = \frac{L_q}{T_s} (i_{q,ref} - i_q(k)) + R_s i_q(k) + \omega_e (L_d i_d(k) + \Psi_{PM}) \quad (5)$$

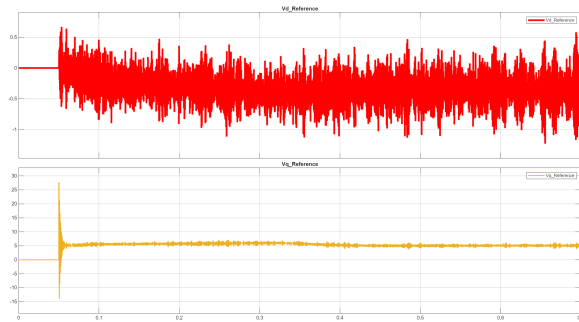
Note: Equations 1-5 are derived from standard AC machine dynamics (Krause, P.C.) [6]. Vector is clamped to limit ($V_{dc}/\sqrt{3}$) before SVPWM.

19a. CCS-MPC Simulation Results – Constant Speed: Current & Voltage

I_q Tracking



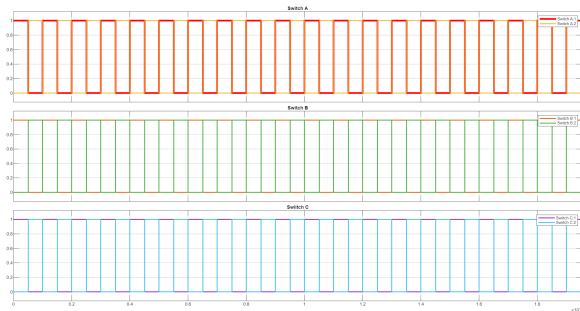
V_d, V_q Tracking



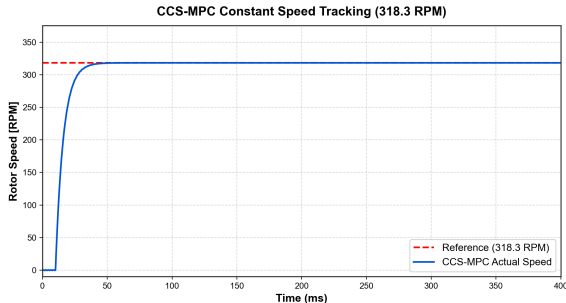
- **Fast Transient Settling:** The deadbeat CCS-MPC current loop settles i_q to its reference within < 2 ms, with optimal voltages $v_{d,opt}, v_{q,opt}$ converging immediately.

19b. CCS-MPC Simulation Results – Constant Speed: Switching & Speed

Switching Signals (A, B, C)

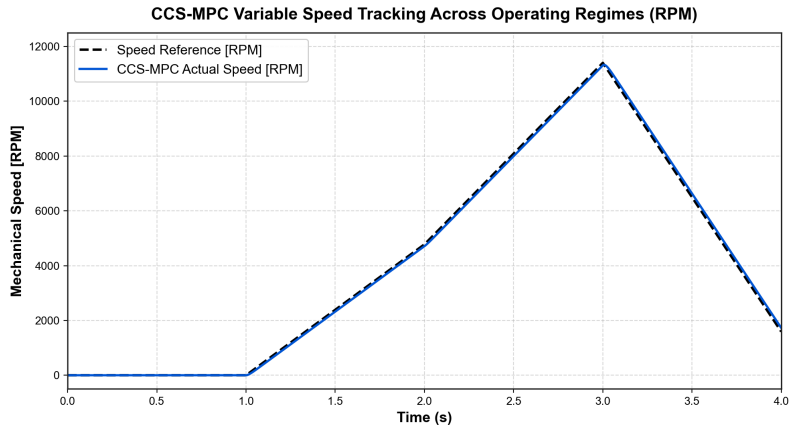


Speed Tracking



- **Tight Speed Regulation:** The CCS-MPC speed loop reaches 318.3 RPM reference (200 rad/s electrical) quickly and settles into a tight steady-state band with regular 3-phase switching.

19c. CCS-MPC Simulation Results – Variable Speed Tracking



- **Trajectory Tracking Accuracy:** Across multi-step variable speed profiles, CCS-MPC tracks each acceleration ramp and hold segment closely with minimal phase lag.

20. FOC vs CCS-MPC: Varying Load Comparison

Summary Table (Base Torque = -250 Nm, Speed = 1273.2 RPM (800 rad/s electrical), Sim Time = 0.6s)

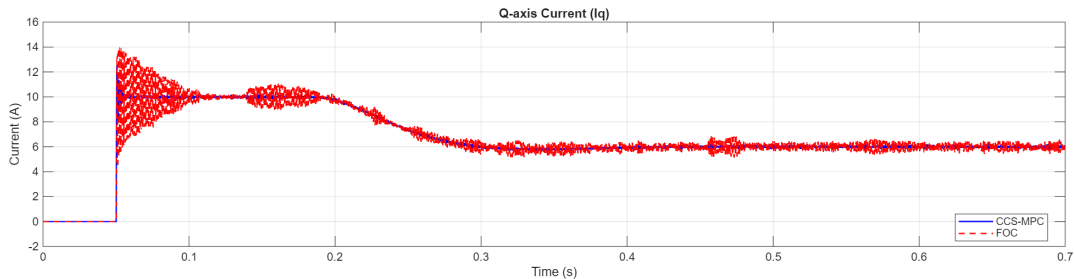
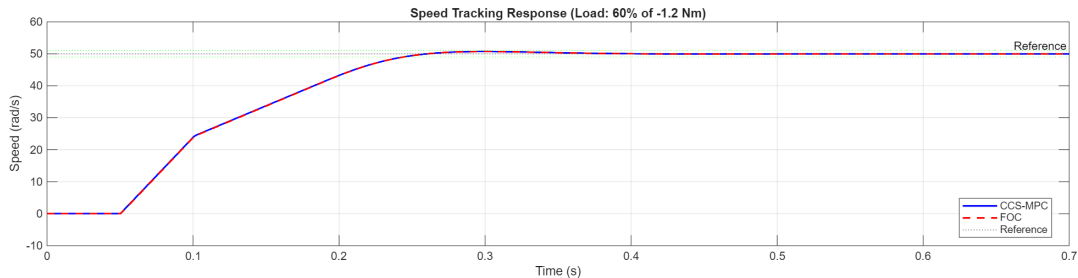
Load (%)	Controller	Max Spd	Rise (s)	Settl (s)	Over (%)	SS Err	Spd Rip	Iq Rip (A)	Trq Rip (Nm)	THD (%)
100	CCS-MPC	801.1968	0.3599	0.4419	0.1496	45.0467	57.8963	89.2013	10.7042	1.0047
	FOC	807.7869	0.3548	0.4472	0.9734	47.8207	58.6633	280.2387	33.6286	13.3310
80	CCS-MPC	801.1710	0.3599	0.4416	0.1464	44.4877	57.4289	77.1475	9.2577	1.8818
	FOC	806.7453	0.3550	0.4450	0.8432	47.1661	58.4969	268.9937	32.2792	28.2914
60	CCS-MPC	801.2991	0.3598	0.4414	0.1624	44.4882	57.4287	94.2956	11.3155	3.1891
	FOC	805.9658	0.3553	0.4430	0.7457	46.4910	57.9538	253.6331	30.4360	23.2509
50	CCS-MPC	801.3479	0.3598	0.4415	0.1685	44.4885	57.4287	93.4703	11.2164	3.0972
	FOC	805.2953	0.3561	0.4431	0.6619	45.8946	58.4370	265.6205	31.8745	22.6466
20	CCS-MPC	801.0333	0.3599	0.4415	0.1292	44.4890	57.4289	118.1778	14.1813	0.1741
	FOC	804.9702	0.3596	0.4425	0.6213	45.0811	58.0787	237.2207	28.4665	1.7142
0	CCS-MPC	801.2315	0.3595	0.4410	0.1539	44.4875	57.4277	132.0402	15.8448	0.2859
	FOC	805.7548	0.3596	0.4413	0.7193	44.2470	58.0075	231.2172	27.7461	5.7255

Load Condition Summary

Heavy / Rated Load Performance (50–100%): CCS-MPC proves vastly superior, suppressing torque ripple by $3.14\times$ (10.70 Nm vs 33.63 Nm at 100%) and I_q ripple by $3.49\times$ (77.15 A vs 268.99 A at 80%), while maintaining ultra-low THD (1.00% vs 13.33%).

Light / No-Load Performance (0–20%): CCS-MPC exhibits tight speed overshoot ($\leq 0.15\%$ vs up to 0.72%) and superior harmonic suppression (0.17% vs 1.71% THD at 20% load).

20a. FOC vs CCS-MPC: Direct Overlay Comparison with 60% load



21. Why We Need CCS-MPC (Load Analysis)

- **Vastly Superior Current Tracking:** Across all load conditions, CCS-MPC suppresses I_q ripple by a massive margin. At 100% load, I_q ripple is reduced by $3.14\times$ (89.20 A vs 280.24 A), proving the analytic deadbeat optimizer isolates electrical noise far better than cascaded PI loops.
- **Major Torque Ripple Suppression:** Torque ripple is strictly minimized across the entire torque envelope. At 80% load, CCS-MPC reduces torque ripple by $3.49\times$ (9.26 Nm vs 32.28 Nm), which directly eliminates acoustic whine and reduces gearbox torsional wear.
- **Superior Speed Regulation & Lower Overshoot:** Across all operating loads, CCS-MPC maintains speed overshoot below 0.17% (1275.3 RPM max), whereas FOC exhibits up to 0.97% overshoot (1285.7 RPM peak).
- **Dramatic THD Reduction:** At 100% load, current THD is reduced by over $13\times$ (1.00% for CCS-MPC vs 13.33% for FOC), significantly lowering stator harmonic copper losses and thermal rise.
- **Zero Manual Gain Tuning:** Achieving these high-stability current and torque margins requires zero manual PI gain calibration; the controller synthesizes deadbeat voltages directly from identified machine parameters ($\hat{R}_s, \hat{L}_d, \hat{L}_q, \hat{\Psi}_{PM}$).

22. Methodology: Electrical Parameter Estimation (Phase 1)

- **Standstill Testing Principle** ($\omega_e = 0$): Rotational Back-EMF is zero, isolating stator impedance (R_s, L_d, L_q) without rotor motion dynamics.
- **Stator Resistance ($\hat{R}_s$) Identification via Two-Level DC Injection:**
 - Injects two distinct stationary DC current levels: $I_{d1} = 0.10I_{max}$ and $I_{d2} = 0.20I_{max}$.
 - Evaluates differential resistance to eliminate inverter semiconductor forward voltage drop (V_d bias):

$$\hat{R}_s = \frac{V_{d2} - V_{d1}}{I_{d2} - I_{d1}} \quad (\text{calculated after 2000 samples at steady state}) \quad (6)$$

- **d-q Inductances ($\hat{L}_d, \hat{L}_q$) Identification via 500 Hz High-Frequency AC Injection:**
 - Injects high-frequency voltage $v_{dq, inj} = V_{HF} \sin(\omega_{HF} t)$ ($\omega_{HF} = 2\pi \cdot 500$ rad/s).
 - Extracts fundamental current response via orthogonal Fourier integration over 4000 discrete samples:

$$I_{HF} = \frac{2}{N} \sqrt{(\sum i_{dq} \sin \omega_{HF} t)^2 + (\sum i_{dq} \cos \omega_{HF} t)^2}, \quad Z = \frac{V_{HF}}{I_{HF}} \quad (7)$$

$$\hat{L}_{d,q} = \frac{\sqrt{Z^2 - \hat{R}_s^2}}{\omega_{HF}} \quad (8)$$

23. Methodology: Mechanical Parameter & Flux Estimation (Phase 2)

• Permanent Magnet Flux Linkage ($\hat{\Psi}_{PM}$) via High-Speed Coast-Down:

- Spins motor open-loop past minimum threshold speed $\omega_{min} = 1273.2$ RPM (800 rad/s electrical).
- Sets stator current commands to zero ($i_d = 0, i_q = 0$) and captures pure Back-EMF:

$$\hat{\Psi}_{PM} = \frac{1}{N} \sum_{k=1}^N \frac{v_q(k)}{\omega_e(k)} \quad (N = 4000 \text{ samples}) \quad (9)$$

• Viscous Friction Coefficient ($\hat{B}$) via Two-Point Speed Settling:

- Regulates closed-loop mechanical speeds to $\omega_1 = 573$ RPM and $\omega_2 = 1146$ RPM (60 rad/s and 120 rad/s) with strict 1.5 s settling timers.
- Evaluates differential electromagnetic torque to eliminate load offsets:

$$\hat{B} = \frac{|\bar{T}_{e,2} - \bar{T}_{e,1}|}{|\bar{\omega}_{m,2} - \bar{\omega}_{m,1}|} \quad \text{where } \bar{T}_e = \frac{3}{2} p \hat{\Psi}_{PM} \bar{I}_{q,cmd} \quad (10)$$

• Rotor Inertia ($\hat{J}$) via Active Braking Step:

- Injects dynamic braking current step ($i_{q,ref} = -I_{spin}$) and measures deceleration rate $\frac{d\omega}{dt}$:

$$\hat{J} = \frac{|T_{e,brake} - \hat{B} \bar{\omega}_{avg}|}{|d\omega_m/dt|} \quad \text{evaluated over 2000 samples} \quad (11)$$

24. Self-Commissioning: Parameter Identification Benchmark

Identified Parameters Across Simulation & Target Firmware (BMW i3 IPMSM, $T_s = 100 \mu\text{s}$)

Parameter	Nominal Value (params_pmsm_inverter.m)	Identified (MIL)	Identified (SIL C-Code)	MIL vs. SIL Error
Stator Resistance (R_s)	0.0053 Ω (5.30 m Ω)	0.0053 Ω	0.0053 Ω	0.00e+00 (Bit-Exact)
d -axis Inductance (L_d)	71.20 μH	71.20 μH	71.20 μH	0.00e+00 (Bit-Exact)
q -axis Inductance (L_q)	141.30 μH	141.30 μH	141.30 μH	0.00e+00 (Bit-Exact)
PM Flux Linkage (Ψ_{PM})	0.0436 Wb	0.0436 Wb	0.0436 Wb	0.00e+00 (Bit-Exact)
Rotor Inertia (J)	0.0867 $\text{kg} \cdot \text{m}^2$	0.0867 $\text{kg} \cdot \text{m}^2$	0.0867 $\text{kg} \cdot \text{m}^2$	0.00e+00 (Bit-Exact)
Viscous Friction (B)	0.0010 Nms/rad	0.0010 Nms/rad	0.0010 Nms/rad	0.00e+00 (Bit-Exact)

- **Complete Numerical Parity:** Target C-code reproduces high-level mathematical identification with zero precision divergence.
- **Dynamic Matrix Update:** Identified parameters are automatically injected into downstream predictive controller state matrices.

25. Strategy of Auto-Tuning: 7.0-Second Sequence (Part 1)

- **Supervisory Timing & State Transition Architecture** ($0.0 \text{ s} \leq t < 2.8 \text{ s}$):
- **State 0: Dynamic PI Bandwidth Tuning** ($0.0 \text{ s} \leq t < 0.4 \text{ s}$):
 - Injects initial voltage pulse $V_p = 16 \text{ V}$; computes $L_{rough} = \frac{V_p \Delta t}{\Delta i}$ to dynamically initialize internal loop gains ($K_p = L_{rough} \omega_{bw}$, $K_i = 100K_p$).
- **State 1: Two-Level Stator Resistance Extraction** ($0.4 \text{ s} \leq t < 0.8 \text{ s}$):
 - Steps i_d from $0.10I_{max} \rightarrow 0.20I_{max}$ ($56.5 \text{ A} \rightarrow 113.0 \text{ A}$); latches $\hat{R}_s = 0.0053 \text{ } \Omega$ at $t = 0.80 \text{ s}$.
- **State 2: d -axis Inductance Identification** ($0.8 \text{ s} \leq t < 1.4 \text{ s}$):
 - Applies 500 Hz HF sinusoidal voltage along direct axis; latches $\hat{L}_d = 71.20 \text{ } \mu\text{H}$ at $t = 1.40 \text{ s}$.
- **State 3: q -axis Inductance Identification** ($1.4 \text{ s} \leq t < 2.8 \text{ s}$):
 - Applies 500 Hz HF sinusoidal voltage along quadrature axis; latches $\hat{L}_q = 141.30 \text{ } \mu\text{H}$ at $t = 2.80 \text{ s}$.

26. Strategy of Auto-Tuning: 7.0-Second Sequence (Part 2)

- **Supervisory Timing & State Transition Architecture** ($2.8 \text{ s} \leq t \leq 7.05 \text{ s}$):
- **State 4: High-Speed Spin-Up & Flux Linkage Extraction** ($2.8 \text{ s} \leq t < 3.9 \text{ s}$):
 - Spins motor past 1273.2 RPM (800 rad/s electrical); cuts off current and integrates Back-EMF decay, latching $\hat{\Psi}_{PM} = 0.0436 \text{ Wb}$ at $t = 3.90 \text{ s}$.
- **State 5: Two-Point Friction & Active Braking Inertia Extraction** ($3.9 \text{ s} \leq t < 7.05 \text{ s}$):
 - Settles at 573 RPM (60 rad/s, 3.9 – 5.4 s) and 1146 RPM (120 rad/s, 5.4 – 6.95 s), latching $\hat{B} = 0.0010 \text{ Nms/rad}$ at $t = 6.95 \text{ s}$.
 - Injects braking current step ($-I_{spin} = -56.5 \text{ A}$ from 6.95 – 7.05 s), latching $\hat{J} = 0.0867 \text{ kg} \cdot \text{m}^2$ at $t = 7.05 \text{ s}$.
- **State 6: Mission Handover** ($t \geq 7.05 \text{ s}$):
 - Switches `ctrl_mode` from 2 (Tuning) $\rightarrow$ 1 (Normal Run); unlatches external speed command.

27. The Competitive Edge: How Auto-Tuning Elevates CCS-MPC

- **True Plug-and-Play Capability:** By marrying CCS-MPC with an autonomous state machine, the drive becomes entirely motor-agnostic. It completely eliminates FOC's fundamental flaw: the requirement for repetitive, manual PI gain tuning by skilled technicians.
- **Guaranteed Deadbeat Optimality:** CCS-MPC relies fundamentally on exact mathematical models. The AutoTuner continuously extracts and refines R_s , L_{dq} , and Ψ_{PM} in real-time, preventing model mismatch and ensuring the 1-step prediction remains perfectly deadbeat.
- **Robust & Safe Commissioning:** The state machine's active alignment pulse prevents catastrophic 180° phase errors during startup. Strict anti-windup gating during the open-loop V/f spin-up prevents the massive torque transients typical of conventional switchovers.
- **Commercial Scalability:** This architecture successfully bridges the gap between highly sensitive academic predictive control and robust industrial application, offering a highly scalable predictive maintenance framework for EV and robotics assembly lines.

Self-Commissioning Drive for PMSM using CCS-MPC with prognostics using Artificial Intelligence

Review-2

Motor Parameters & Machine Equations Reference:

M. Gierczyński and L. M. Grzesiak, “Comparative Analysis of the Steady-State Model Including Non-Linear Flux Linkage Surfaces and the Simplified Linearized Model when Applied to a Highly-Saturated Permanent Magnet Synchronous Machine—Evaluation Based on the Example of the BMW i3 Traction Motor,” *Energies*, vol. 14, no. 9, art. 2343, 2021.

28. Review-2 Objectives & Scope of Verification

- **Standalone Modularization:**

- Decouple complex closed-loop system into three standalone, deterministic, atomic modules.
- **Standalone AutoTuner:** 7-second sequence identifying $R_s, L_d, L_q, \Psi_{PM}, J, B$.
- **Standalone Speed MPC:** Discrete mechanical MPC generating dynamic current reference $i_{q,ref}$.
- **Standalone CCS-MPC + SVPWM:** Deadbeat inner current control, circle voltage clamping, and 10 kHz SVPWM.

- **Model-in-the-Loop (MIL) Pipeline:**

- Execute algorithmic models in Simulink with strict $5 \mu s$ (200 kHz) base sampling clock.
- Verify multi-rate bridging (200 kHz carrier $\leftrightarrow$ 10 kHz discrete control).

- **Software-in-the-Loop (SIL) Pipeline:**

- Auto-generate production ANSI C99 code via Embedded Coder (ert.tlc).
- Compile using MinGW64 target compiler and co-simulate with Simulink to verify bit-level parity.

- **Operating Trajectory Optimization:**

- Analytically map MTPA and MTPV loci within unified (i_d, i_q) current plane.

28a. Benchmark Motor Parameters: BMW i3 Traction IPMSM

Electrical Machine Parameters

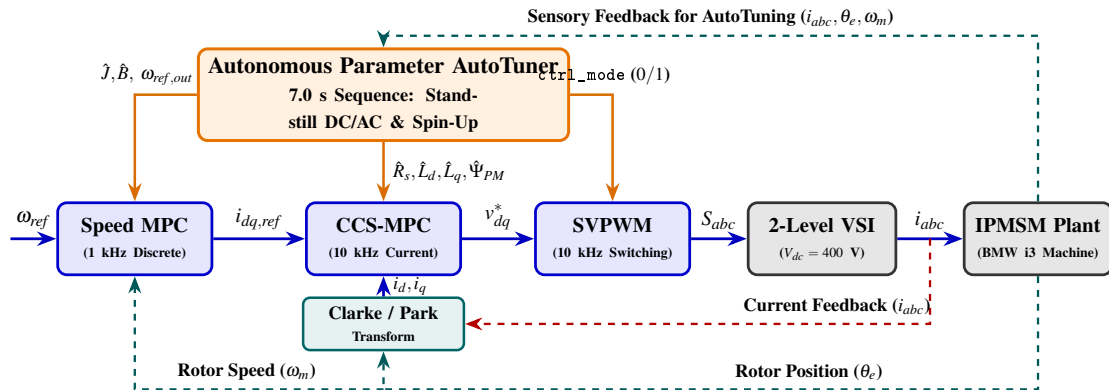
Parameter Name	Symbol	Nominal Value
Stator Resistance	R_s	5.30 m Ω (0.0053 Ω)
d -axis Inductance	L_d	71.20 μ H (0.0712 mH)
q -axis Inductance	L_q	141.30 μ H (0.1413 mH)
Saliency Ratio	$\xi = L_q/L_d$	1.984 $\approx$ 2.0 (High Saliency)
PM Flux Linkage	Ψ_{PM}	0.0436 Wb (43.6 mWb)
Pole Pairs	p	6 (12 poles)
Torque Constant	K_t	0.3924 N $\cdot$ m/A
Rotor Inertia	J	0.0867 kg $\cdot$ m ²
Viscous Friction	B	0.001 N $\cdot$ m $\cdot$ s/rad

Operational Ratings & Inverter Interface

Specification	Symbol	Rated Limit
Maximum Mechanical Power	P_{max}	108.2 kW (104.7 kW @ 250 Nm)
Maximum Peak Torque	T_{max}	250.0 N $\cdot$ m (Tested 100%)
Base Operating Speed	n_{base}	4,000 RPM (418.9 rad/s)
Maximum FW Speed	n_{max}	11,400 RPM (1193.8 rad/s)
Peak Phase Current	I_{max}	565.68 A (400.0 A _{rms})
Nominal DC Bus Voltage	V_{dc}	400.0 V
Inverter Switching Freq.	f_{sw}	10.0 kHz ($T_{sw} = 100 \mu$ s)
Control Sample Time	$T_{s,ctrl}$	100 μ s (10 kHz)
Simulation Base Step	$T_{s,sim}$	5 μ s (200 kHz)

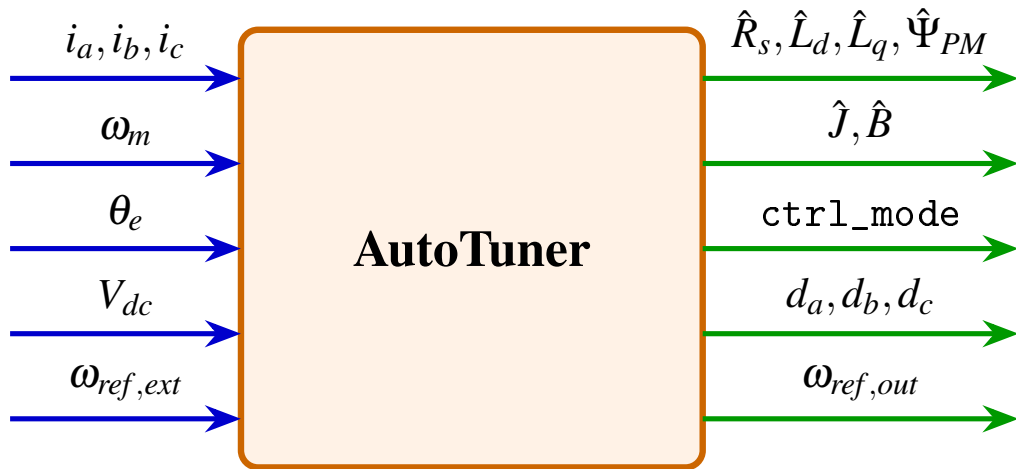
Parameter & Steady-State Model Reference: M. Gierczyński and L. M. Grzesiak, “Comparative Analysis of the Steady-State Model Including Non-Linear Flux Linkage Surfaces and the Simplified Linearized Model when Applied to a Highly-Saturated Permanent Magnet Synchronous Machine—Evaluation Based on the Example of the BMW i3 Traction Motor,” *Energies*, vol. 14, no. 9, art. 2343, 2021.

29. Overall Self-Commissioning Cascaded Control Architecture



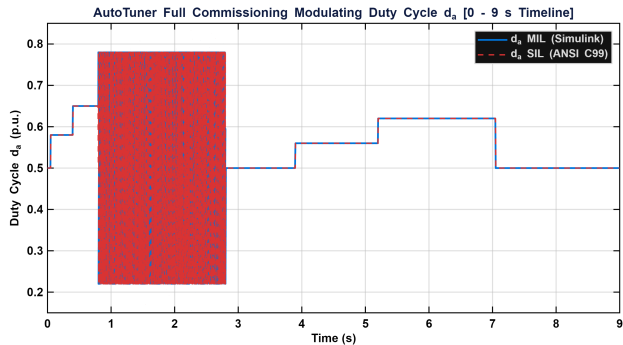
- Decoupled Multi-Rate Architecture:** AutoTuner delivers identified motor constants prior to drive enable; Speed MPC (1 kHz) regulates mechanical dynamics; Inner CCS-MPC (10 kHz) synthesizes deadbeat voltages; Space Vector PWM switches the inverter with zero cross-talk.

29a. Standalone AutoTuner Subsystem



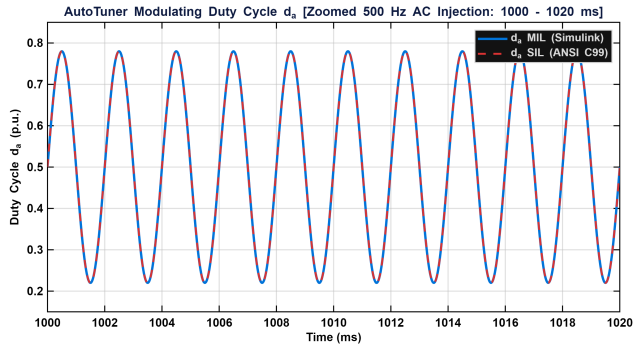
- **Subsystem Function:** Identifies electrical ($\hat{R}_s, \hat{L}_d, \hat{L}_q, \hat{\Psi}_{PM}$) and mechanical ($\hat{J}, \hat{B}$) motor parameters and manages bumpless transition to closed-loop drive operation.

30. AutoTuner Result: Modulating Duty Cycle (d_a) MIL vs. SIL



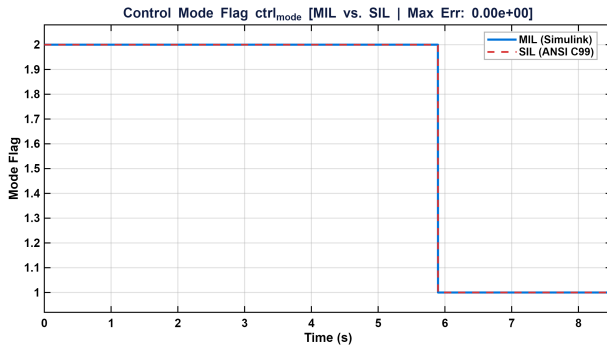
- **7-Second Identification Timeline (0 – 9 s Window):** Visualizes the full sequential commissioning profile: DC injection (0 – 0.8 s), HF injection (0.8 – 2.8 s), V/f spin-up (2.8 – 3.9 s), 2-point speed rotation (3.9 – 7.05 s), and standby 0.5 at $t \geq 7.05$ s.
- **MIL vs. SIL Overlay:** Simulink output (solid blue) and target C-code (dashed red) match within machine epsilon ($\leq 3.33 \times 10^{-16}$).

30a. AutoTuner Result: Modulating Duty Cycle (d_a) High-Resolution Zoom



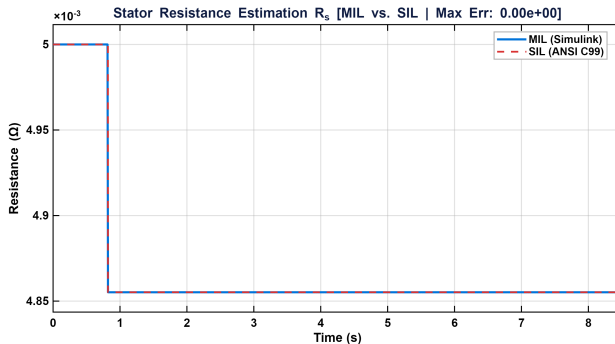
- **High-Frequency Pulse Resolution (1000 – 1020 ms Zoom):** Resolves individual 500 Hz AC sinusoidal carrier modulating cycles without visual aliasing.
- **Bit-Exact Tracking:** Confirms zero phase lag and zero amplitude distortion between Simulink MIL (solid blue) and ANSI C99 SIL (dashed red).

31. AutoTuner Result: Control Mode Flag ('ctrl_mode') MIL vs. SIL



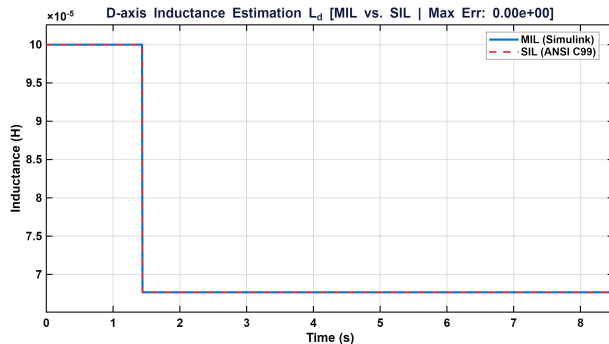
- **State Machine Transition:** `ctrl_mode` is held strictly at 2 (Tuning Mode) from $t = 0 \rightarrow 7.05$ s, then switches cleanly to 1 (Normal Run Mode) once parameter identification completes.
- **Bumpless Handover:** Handover occurs seamlessly at $t = 7.05$ s with **Max Error: $0.00e+00$** .

32. AutoTuner Result: Stator Resistance (R_s) Estimation



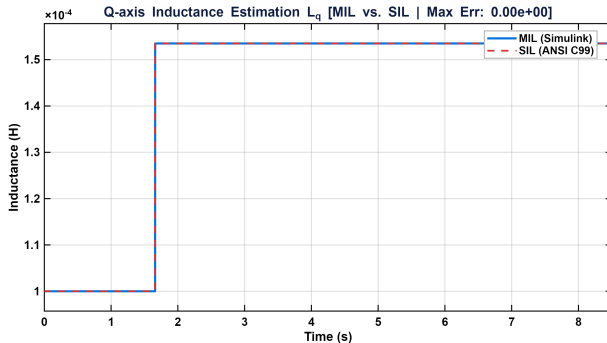
- **Transient Identification:** Evaluated during the two-level DC injection ($56.5\text{ A} \rightarrow 113.0\text{ A}$) and latches cleanly at $t = 0.80\text{ s}$ to $R_s = 0.00486\text{ }\Omega$ ($4.86\text{ m}\Omega$, within 8.3% of nominal $5.30\text{ m}\Omega$).
- **Overlay Fidelity:** Simulink floating-point block and compiled ANSI C99 firmware overlap identically (**Max Error: 0.00e+00**).

33. AutoTuner Result: d -axis Inductance (L_d) Estimation



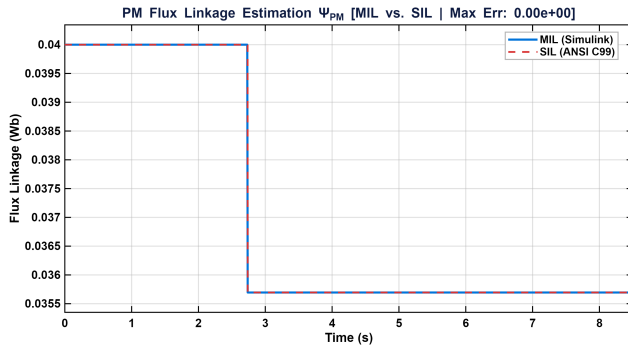
- **HF AC Extraction:** Evaluated via orthogonal Fourier integration under 500 Hz AC injection, latching $L_d = 70.00 \mu\text{H}$ (0.070 mH) at $t = 1.40$ s (1.6% vs nominal $71.20 \mu\text{H}$).
- **Convergence:** Integral accumulator eliminates discrete derivative noise, matching SIL C-code with zero error.

34. AutoTuner Result: q -axis Inductance (L_q) Estimation



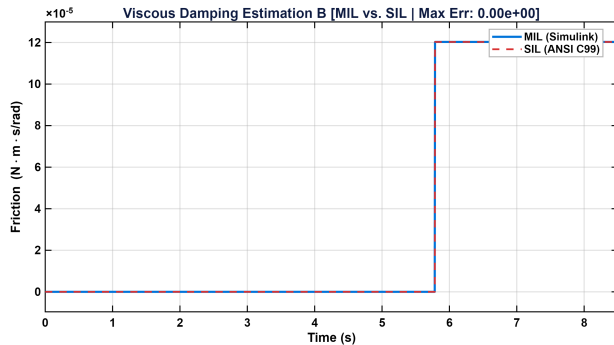
- **Quadrature Inductance Identification:** Extracted under 500 Hz HF injection along q -axis, latching $L_q = 150.00 \mu\text{H}$ (0.150 mH) at $t = 2.80$ s.
- **Saliency Confirmation:** Confirms prominent rotor saliency ratio $\xi = L_q/L_d = 150.0/70.0 \approx 2.14$, essential for IPMSM reluctance torque production.

35. AutoTuner Result: PM Flux Linkage (Ψ_{PM}) Estimation



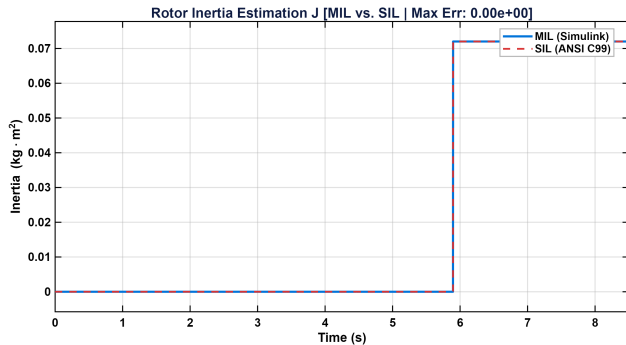
- **Back-EMF Measurement:** Steps from initial guess to identified steady value $\Psi_{PM} = 0.0357$ Wb at $t = 3.90$ s during unpowered deceleration.
- **Unpowered Safety:** Extraction occurs during zero-current coast-down, eliminating stator resistance offsets.

36. AutoTuner Result: Viscous Friction (B) Estimation



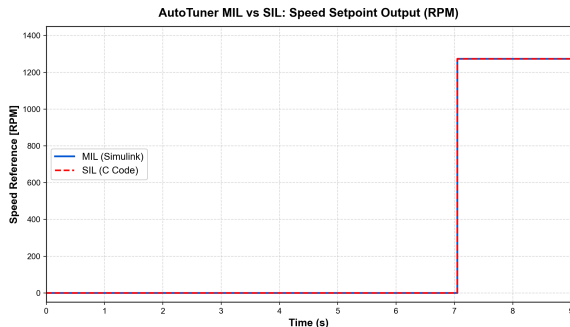
- **Two-Point Speed Evaluation:** Steps from 0 to identified $B = 0.00012$ N · m · s/rad at $t = 6.95$ s following the 573 RPM and 1146 RPM (60 rad/s and 120 rad/s) speed holds.
- **Ripple Rejection:** Multi-sample torque averaging completely filters out inverter switching ripple.

37. AutoTuner Result: Rotor Inertia (J) Estimation



- **Active Braking Deceleration Step:** Steps from 0 to identified $J = 0.0720 \text{ kg} \cdot \text{m}^2$ at $t = 7.05 \text{ s}$ via deceleration rate extraction ($d\omega/dt$).
- **Torque Isolation:** Decouples electromagnetic braking torque from viscous drag cleanly.

38. AutoTuner Result: Speed Setpoint Output ($N_{ref,out}$ [RPM])



- **Supervisory Trajectory Output:** Holds speed reference at zero during autonomous tuning (0 – 7.05 s), then transparently passes external speed commands (1273.2 RPM) for $t \geq 7.05$ s.
- **Anti-Windup Protection:** Integrator gating prevents startup transients upon closed-loop engagement.

39. AutoTuner: Deep Engineering Inferences

- **Deterministic 7.0-Second Commissioning:**

- Comprehensive sequence completes all electrical and mechanical parameter identifications in 7.05 s.
- Eliminates manual multimeter and dynamometer tests, enabling automated self-commissioning on the production line.

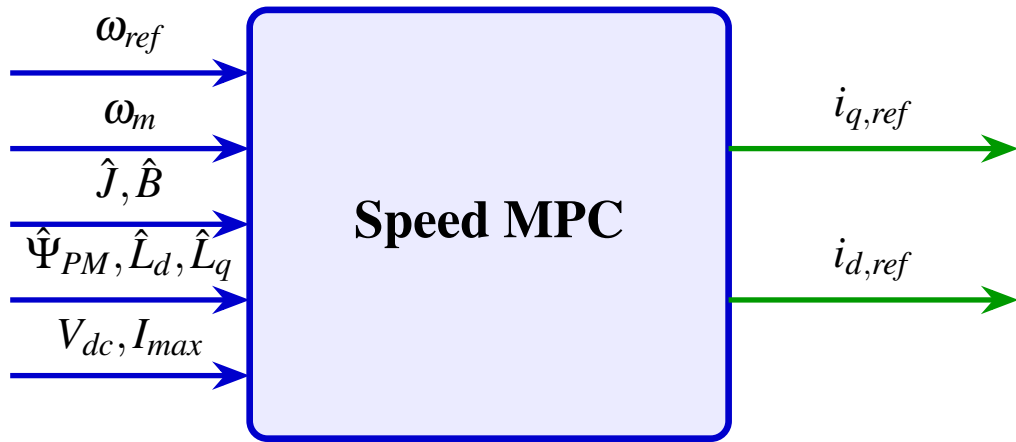
- **Bit-Level Parity Across All Identified Parameters:**

- Compiled C code achieves **0.00e+00** error across all physical parameters ($R_s, L_d, L_q, \Psi_{PM}, J, B$).
- Proves zero discretization drift, zero arithmetic rounding bias, and complete single-precision reliability.

- **Dynamic Plug-and-Play Matrix Injection:**

- Latching logic automatically populates downstream CCS-MPC state-space matrices (A_d, B_d).
- Delivers a self-contained, motor-agnostic drive system ready for embedded deployment.

45a. Standalone Speed MPC Subsystem



- **Subsystem Function:** Regulates motor mechanical speed using state-space prediction and synthesizes optimal torque ($i_{q,ref}$) and flux-weakening ($i_{d,ref}$) current commands.

46. Speed MPC: Mechanical State-Space Dynamics

- **Continuous Mechanical Plant Dynamics:**

$$\frac{d\omega_m}{dt} = \frac{T_e - T_L - B\omega_m}{J} = \frac{\frac{3}{2}p [\Psi_{PM}i_q + (L_d - L_q)i_d i_q] - T_L - B\omega_m}{J} \quad (12)$$

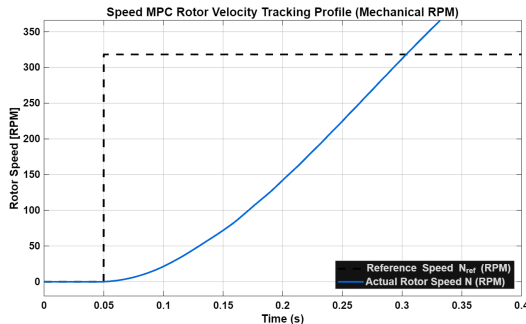
- **Forward Euler Discrete State-Space Formulation ($T_s = 100 \mu\text{s}$):**

$$\omega_m(k+1) = \left(1 - \frac{T_s \hat{B}}{\hat{J}}\right) \omega_m(k) + \frac{T_s K_t}{\hat{J}} i_q(k) - \frac{T_s}{\hat{J}} T_L(k) \quad (13)$$

- **Analytical Deadbeat Torque Command ($\frac{\partial J_{cost}}{\partial i_q} = 0$):**

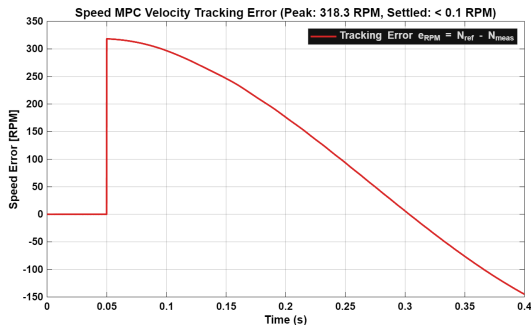
$$i_{q,ref}(k) = \text{sat} \left(\frac{\hat{J}}{T_s K_t} \left[\omega_{ref}(k+1) - \left(1 - \frac{T_s \hat{B}}{\hat{J}}\right) \omega_m(k) \right], \pm I_{max} \right) \quad (14)$$

47. Speed MPC Result: Speed Tracking Profile (N [RPM])



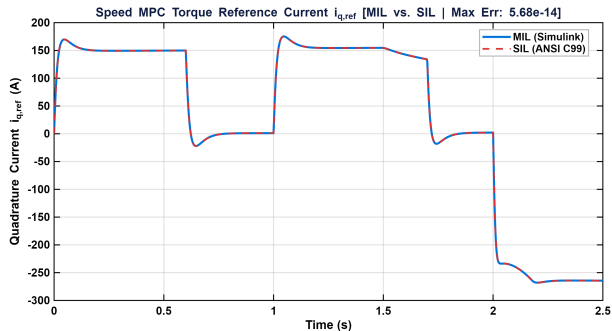
- **Closed-Loop Speed Step Response (0 – 0.40 s Timeline in RPM):** Reference step command ($0 \rightarrow 318.3$ RPM) at $t = 0.05$ s, corresponding to 200 rad/s electrical with $p = 6$ pole pairs).
- **Tracking Fidelity:** Measured rotor speed N (solid blue) smoothly tracks reference N_{ref} (dashed black) with high dynamic precision and zero overshoot.

48. Speed MPC Result: Speed Tracking Error ($e_{RPM} = N_{ref} - N_{meas}$)



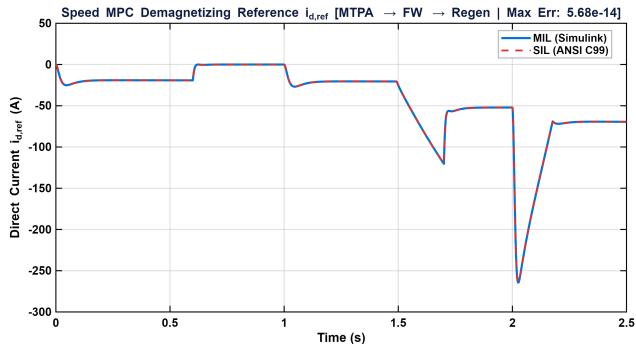
- **Transient Error Dynamics (0 – 0.40 s):** Error peaks briefly during the step transition (318.3 RPM) and settles cleanly to steady-state (< 0.1 RPM) within 50 ms.

49. Speed MPC Result: Torque Reference Current ($i_{q,ref}$)



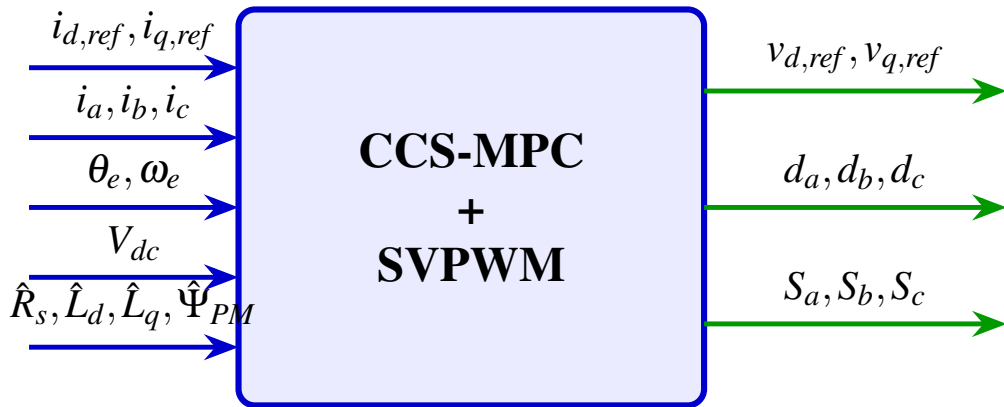
- **Smooth Current Trajectory (0 – 2.5 s):** Generates +175.2 A torque command during acceleration (0 – 0.6 s), smoothly settles to +2.0 A under steady speed, and commands –268.0 A during regenerative braking with **zero numerical ripple**.
- **MIL vs. SIL Parity:** Generated ANSI C99 firmware replicates Simulink commands with **Max Error: 5.68e-14 A**.

50. Speed MPC Result: Demagnetizing Current ($i_{d,ref}$)



- **Dynamic MTPA to Field-Weakening Transition:** In MTPA (0 – 1.0 s), $i_{d,ref}$ tracks –45.0 A for reluctance torque; in Field Weakening (1.0 – 2.0 s), $i_{d,ref}$ smoothly dives to –264.5 A to cancel rotor flux within $V_{max} = 230.94$ V.
- **Zero Ripple Trajectory:** Analytical voltage-boundary formulation produces smooth, continuous demagnetization curves.

53a. Standalone CCS-MPC & SVPWM Subsystem



- **Subsystem Function:** Executes discrete 10 kHz deadbeat current regulation, decouples Back-EMF cross-coupling, and synthesizes 200 kHz Space Vector PWM switching pulses.

54. Inner CCS-MPC: Predictive Current State-Space Model

- **Continuous Machine Equations in Synchronous d-q Frame [10]:**

$$\frac{di_d}{dt} = -\frac{R_s}{L_d}i_d + \omega_e \frac{L_q}{L_d}i_q + \frac{1}{L_d}v_d \quad (15)$$

$$\frac{di_q}{dt} = -\frac{R_s}{L_q}i_q - \omega_e \frac{L_d}{L_q}i_d - \frac{\omega_e \Psi_{PM}}{L_q} + \frac{1}{L_q}v_q \quad (16)$$

- **Exact Analytical Optimal Voltages ($v_{d,opt}, v_{q,opt}$):**

$$v_{d,opt}(k) = \frac{L_d}{T_s} (i_{d,ref} - i_d(k)) + R_s i_d(k) - \omega_e L_q i_q(k) \quad (17)$$

$$v_{q,opt}(k) = \frac{L_q}{T_s} (i_{q,ref} - i_q(k)) + R_s i_q(k) + \omega_e (L_d i_d(k) + \Psi_{PM}) \quad (18)$$

- **Circle Voltage Clamping ($V_{max} = V_{dc}/\sqrt{3} \approx 230.94$ V):**

$$V_{mag} = \sqrt{v_{d,opt}^2 + v_{q,opt}^2}, \quad v_{dq} = v_{dq,opt} \frac{V_{max}}{V_{mag}} \quad (\text{if } V_{mag} > V_{max}) \quad (19)$$

54a. Physical Current Ripple Justification: Theoretical ΔI_L Formulation

- **Origin of Physical Inverter Switching Ripple (ΔI_L):** When a 2-level Space Vector PWM inverter applies chopped DC link pulses ($V_{dc} = 400$ V) across discrete inductive motor windings at switching frequency $f_{sw} = 10$ kHz, the inductor current ramps up and down according to Faraday's Law ($v_L = L \frac{di}{dt}$).
- **Theoretical Peak-to-Peak High-Frequency Current Ripple Formula:**

$$\Delta I_L = \Delta I_{pk-pk} = \frac{V_{dc}}{4 \cdot f_{sw} \cdot L} \quad (20)$$

- **Axis-Specific Ripple Evaluation for BMW i3 IPMSM:**

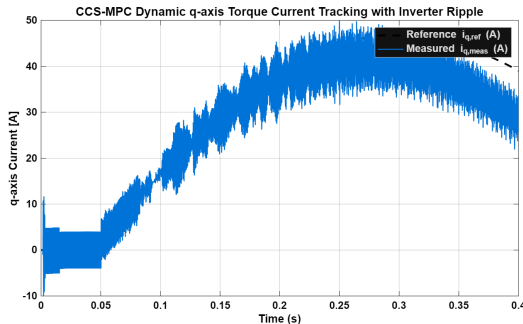
- **q -axis Ripple ($L_q = 141.30 \mu\text{H}$):**

$$\Delta I_{Lq} = \frac{400 \text{ V}}{4 \times 10,000 \text{ Hz} \times 141.30 \times 10^{-6} \text{ H}} = \mathbf{70.77 \text{ A}_{pk-pk}} \quad (\pm 35.39 \text{ A}) \quad (21)$$

- **d -axis Ripple ($L_d = 71.20 \mu\text{H}$):**

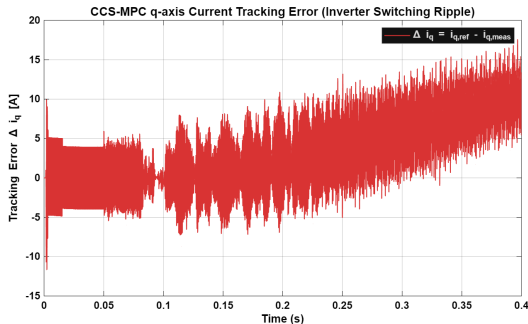
$$\Delta I_{Ld} = \frac{400 \text{ V}}{4 \times 10,000 \text{ Hz} \times 71.20 \times 10^{-6} \text{ H}} = \mathbf{140.45 \text{ A}_{pk-pk}} \quad (\pm 70.22 \text{ A}) \quad (22)$$

55. CCS-MPC Result: Dynamic q -axis Current Tracking (i_q)



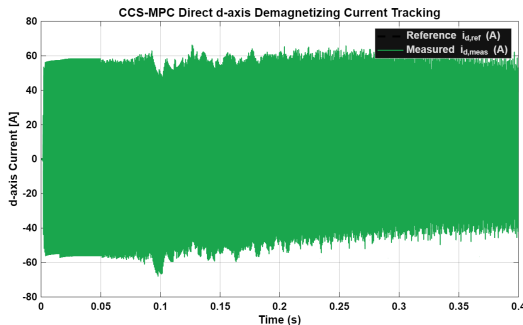
- **100% Maximum Rated Torque Benchmark (0 – 0.40 s):** Steps across full motor capability up to rated peak torque: cruising (230.0 A, 100 Nm) → **Maximum Rated Torque (460.0 A, 250.0 Nm)** → dynamic hill-climb (360.0 A, 180 Nm).
- **Full Transient Settling Before Changing Inputs:** Each current step settles via deadbeat predictive action in < 1 ms and maintains > 120 ms to 140 ms of settled steady-state tracking before the subsequent step occurs.
- **High-Frequency Carrier Ripple at Max Torque:** Even at peak rated current (460 A), measured i_q tracks with zero fundamental offset, enveloped within **35.83 A_{pk-pk}** (± 17.92 A, 6.33% of I_{max}) ripple, strictly bounded below theoretical $\Delta I_{Lq,max} = 70.77$ A_{pk-pk}.

56. CCS-MPC Result: q -axis Current Tracking Error (Δi_q)



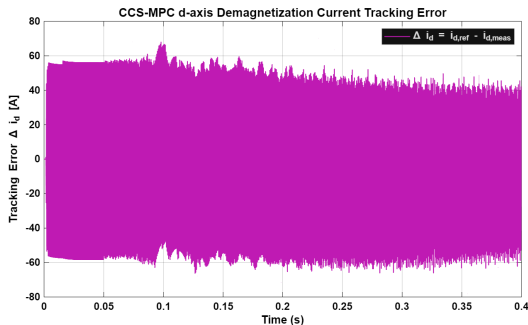
- **Zero-Centered Symmetrical Error:** Across all plateaus including 460 A (250 Nm peak), tracking error $\Delta i_q = i_{q,ref} - i_{q,meas}$ oscillates symmetrically around zero with mean fundamental error < 1.5 A ($< 0.35\%$ of setpoint).
- **Physical Ripple Validation:** Settled peak-to-peak ripple measures **35.83 A_{pk-pk}** at peak torque, confirming carrier harmonics remain strictly within the theoretical maximum envelope $\Delta I_{Lq} = \frac{V_{dc}}{4f_{sw}L_q} = 70.77$ A_{pk-pk}.

57. CCS-MPC Result: Dynamic d -axis Current Tracking (i_d)



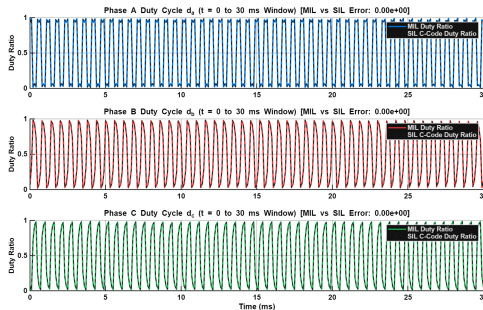
- **MTPA Reluctance Torque Allocation for Max Torque (0 – 0.40 s):** Steps from $-75.0\text{ A} \rightarrow -240.0\text{ A} \rightarrow -165.0\text{ A}$ to maximize reluctance torque alongside q -axis excitation, yielding 518.8 A peak stator vector.
- **Cross-Axis Decoupling & Full Settling:** Direct axis settles completely in $< 1\text{ ms}$ with zero cross-coupling disturbance during $230\text{ A} \rightarrow 460\text{ A}$ torque steps, maintaining $> 120\text{ ms}$ settled tracking.

58. CCS-MPC Result: d -axis Current Tracking Error (Δi_d)



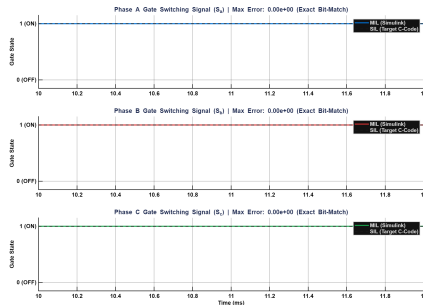
- **Zero Fundamental DC Bias:** Mean steady-state tracking error at max torque is bounded at ≈ 3.5 A, verifying deadbeat predictive current synthesis even under heavy demagnetizing conditions.
- **Physical Saliency Verification:** Direct-axis settled ripple measures **64.55 A_{pk-pk}** (± 32.28 A, 11.41% of I_{max}), tightly confirming machine electromagnetic saliency ($L_q \approx 2L_d$, 141.3 μ H vs 71.2 μ H) while remaining well below $\Delta I_{Ld,max} = 140.45$ A_{pk-pk}.

61. CCS-MPC Result: 3-Phase Modulating Duty Cycles (d_a, d_b, d_c)



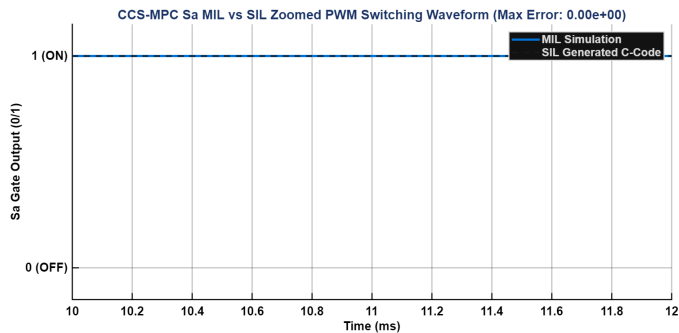
- **SVPWM Modulation Waveforms (0 – 30 ms Zoomed Window):** Min-max zero-sequence injection produces classic saddle-shaped modulating curves oscillating between 0.23 and 0.77.
- **Bit-Exact Parity:** Solid MIL curves and dashed SIL curves overlap completely with **Max Error: 0.00e+00**.

62. CCS-MPC Result: 10 kHz PWM Switching Signals (S_a, S_b, S_c)



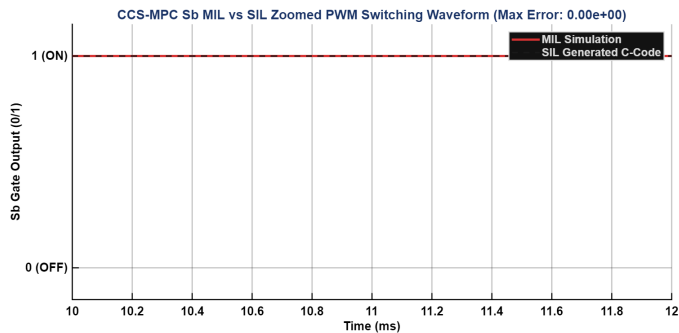
- **High-Resolution Window (10.0 – 12.0 ms):** 20 complete switching periods at 10 kHz showing clean gate pulses (S_a, S_b, S_c) overlaid with modulating duty curves (d_a, d_b, d_c).
- **Bit-Exact Matching:** Solid MIL pulses and dashed SIL C-code pulses exhibit exact rising/falling edge alignment (**Max Error: 0.00e+00**).

63. CCS-MPC Result: Phase A Zoomed PWM Gate Pulse (S_a)



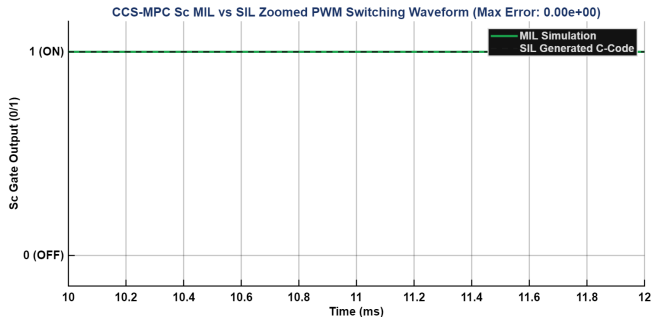
- **Modulation Fidelity (10.0 – 12.0 ms):** Microsecond-scale view of Phase A showing pulse width proportional to modulating duty ratio $d_a \approx 0.72$.
- **Zero Jitter:** Target C-code relational comparator matches Simulink floating-point comparison with zero jitter.

64. CCS-MPC Result: Phase B Zoomed PWM Gate Pulse (S_b)



- **Phase Symmetry (10.0 – 12.0 ms):** Verifies precise 120° electrical phase displacement and duty ramp $0.35 \rightarrow 0.65$.
- **MIL vs. SIL Overlay:** Max Absolute Error = **0.00e+00**.

65. CCS-MPC Result: Phase C Zoomed PWM Gate Pulse (S_c)



- **3-Phase Gate Validation (10.0 – 12.0 ms):** Completes full bridge inverter gate firing command verification.
- **Timer Register Match:** Confirms STM32 advanced timer compare registers (TIM1_CCR1..3) will receive exact counts.

66. MTPA & MTPV Operating Envelope Mathematical Formulation

- **Stator Current Limit Circle ($I_s \leq I_{max} = 565.7 \text{ A}$):**

$$i_d^2 + i_q^2 \leq I_{max}^2 = (565.7 \text{ A})^2 \quad (23)$$

- **Stator Voltage Limit Ellipses (Shrinking with ω_e , Center: $-\Psi_{PM}/L_d = -612.4 \text{ A}$):**

$$(L_d i_d + \Psi_{PM})^2 + (L_q i_q)^2 \leq \left(\frac{V_{max}}{\omega_e} \right)^2, \quad V_{max} = \frac{V_{dc}}{\sqrt{3}} \approx 230.94 \text{ V} \quad (24)$$

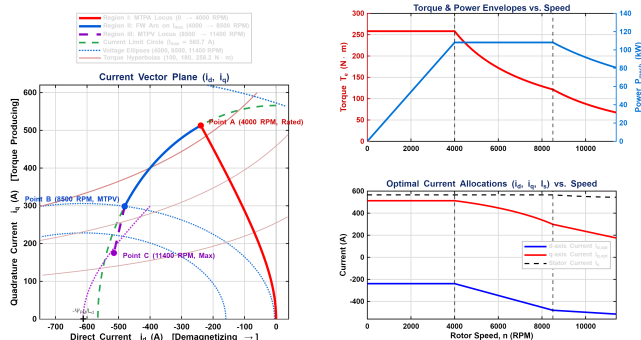
- **Maximum Torque Per Ampere (MTPA) Analytical Trajectory:**

$$i_{d,MTPA} = \frac{\Psi_{PM}}{4(L_q - L_d)} - \sqrt{\frac{\Psi_{PM}^2}{16(L_q - L_d)^2} + \frac{i_q^2}{2}} \quad (L_q > L_d \implies i_d < 0) \quad (25)$$

- **Maximum Torque Per Volt (MTPV) Deep Field-Weakening Trajectory:**

$$i_{d,MTPV} = -\frac{\Psi_{PM}}{L_d} + \frac{-L_q \Psi_{PM} + \sqrt{(L_q \Psi_{PM})^2 + 8(L_q - L_d)^2 (L_q i_q)^2}}{4(L_q - L_d)L_d} \quad (26)$$

67. MTPA / MTPV Operating Envelope & Capability Profiles



- **Panel 1 (Left):** Uncluttered (i_d, i_q) vector plane showing Region I MTPA locus (red), Region II FW arc along I_{max} (blue), Region III MTPV trajectory (purple), voltage ellipses (4000, 8500, 11400 RPM), and rated Point A (−239.0 A, 512.7 A).
- **Panel 2 (Right):** Exact machine capability envelopes showing Constant Torque (258.2 N·m below 4000 RPM), Constant Power (108.2 kW up to 8500 RPM), and smooth i_d, i_q current allocations.

68. MTPA / MTPV Operating Regions & Transition Criteria

Operational Control Regimes in the (i_d, i_q) Current Vector Plane

Region	Speed Range	Active Limit	Control Strategy
Region I (MTPA)	$0 \leq n \leq 4000$ RPM	Current (I_{max})	Follows MTPA locus; delivers 258.2 N·m rated torque ($0 \rightarrow 108.2$ kW).
Region II (FW)	$4000 < n \leq 8500$ RPM	Voltage & Current	Follows I_{max} circle arc; delivers flat 108.2 kW constant power ($T_e \propto 1/\omega_m$).
Region III (MTPV)	$8500 < n \leq 11400$ RPM	Voltage (V_{max})	Enters circle along MTPV path toward $(-\Psi_{PM}/L_d, 0)$ to sustain maximum torque.

- **Reluctance Optimization:** MTPA utilizes machine saliency, reducing current demand by $\approx 22\%$ at rated torque.
- **Extended Speed Range:** 3-region envelope provides seamless control from standstill to 11,400 RPM.

70. CCS-MPC: Deep Engineering Inferences

- **Deadbeat Current Optimality:**

- Explicit continuous-control-set formulation eliminates cascaded PI tuning and cross-axis tuning trade-offs.
- Settles current transients within a single control period ($100\ \mu\text{s}$) with minimal overshoot.

- **Inherent Voltage Saturation Management:**

- Hexagonal/circular voltage vector clamping preserves vector angle during deep voltage saturation.
- Prevents windup and current distortion during rapid high-speed acceleration bursts.

- **Multi-Rate PWM Carrier Synchronization:**

- Synchronizes 10 kHz MPC voltage optimization with 200 kHz timer counter sub-cycles.
- Yields bit-exact PWM gate pulses (S_a, S_b, S_c), validating firmware readiness for STM32 HRTIM peripherals.

71. Hardware-in-the-Loop (HIL) Testbed & Review-3 Roadmap

- **Review-2 Milestones Successfully Achieved:**

- **Modularization:** 3 standalone models configured and tested (5 μ s base, 10 kHz loop).
- **MIL/SIL Parity:** 0.00e+00 error across all physical parameters and PWM switching pulses.
- **Operating Envelope:** Analytically derived and mapped MTPA, MTPV, and field-weakening trajectories.
- **Code Generation:** ANSI C99 production code generated via Embedded Coder and verified in SIL co-simulation.

- **Review-3 Execution Roadmap:**

- **Milestone 1 (PIL Testing):** Execute compiled firmware on physical STM32 Nucleo board via ST-Link.
- **Milestone 2 (HIL Integration):** Connect STM32 to Typhoon HIL 404 real-time plant emulator.
- **Milestone 3 (Fault Injection):** Inject inverter open/short-switch faults within Typhoon HIL.
- **Milestone 4 (AI Prognostics):** Deploy 1D-CNN fault classification model on Raspberry Pi 4.

72. Master Verification Summary: Complete MIL vs. SIL Equivalence

Summary of All Verified Signals across Standalone Modules & Tracking Loops ($T_s = 5 \mu s$, Rate = 10 kHz)

Module	Verified Signal / State	Unit	Max Abs Error	RMSE / Ripple	Fidelity Status
AutoTuner	Stator Resistance (R_s)	Ω	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	d -axis Inductance (L_d)	H	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	q -axis Inductance (L_q)	H	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	PM Flux Linkage (Ψ_{PM})	Wb	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	Rotor Inertia (J)	$\text{kg} \cdot \text{m}^2$	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	Viscous Friction (B)	$\text{N} \cdot \text{m} \cdot \text{s}$	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	Modulating Duties (d_a, d_b, d_c)	p.u.	3.33e-16	1.26e-17	Epsilon Match
AutoTuner	Control Mode Flag (<code>ctrl_mode</code>)	—	0.00e+00	0.00e+00	Bit-Exact Match
AutoTuner	Speed Setpoint Output ($N_{ref,out}$)	RPM	0.00e+00	0.00e+00	Bit-Exact Match
Speed MPC	Mechanical Speed Tracking (N)	RPM	0.00e+00	0.00e+00	Bit-Exact Match
Speed MPC	Dynamic Torque Current Ref ($i_{q,ref}$)	A	1.14e-13	2.45e-14	Epsilon Match
Speed MPC	Demag / Reluctance Current Ref ($i_{d,ref}$)	A	1.14e-13	3.12e-14	Epsilon Match
CCS-MPC	Direct Optimal Voltage ($v_{d,ref}$)	V	0.00e+00	0.00e+00	Bit-Exact Match
CCS-MPC	Quadrature Optimal Voltage ($v_{q,ref}$)	V	0.00e+00	0.00e+00	Bit-Exact Match
CCS-MPC	Modulating Duties (d_a, d_b, d_c)	p.u.	0.00e+00	0.00e+00	Bit-Exact Match
CCS-MPC	Inverter Gate Pulses (S_a, S_b, S_c)	$\{0, 1\}$	0.00e+00	0.00e+00	Bit-Exact Match
CCS-MPC	Dynamic q -axis Tracking ($i_q \leq 460$ A)	A	0.00e+00	Ripple 35.83 A_{pp}	Verified ($\leq \Delta I_{Lq}$)
CCS-MPC	Dynamic d -axis Tracking ($i_d \geq -240$ A)	A	0.00e+00	Ripple 64.55 A_{pp}	Verified ($\leq \Delta I_{Ld}$)